

4.8 L'Hopital's Rule; Indeterminate Forms

Motivation: How could we evaluate the following limits?

$$\lim_{x \rightarrow 1} \frac{\ln(x)}{x-1} \qquad \lim_{x \rightarrow \infty} \frac{\ln(x)}{x-1}$$

The idea we will discuss in this section will help us evaluate limits that have an indeterminate form.

Definition (Intuitive): When direct substitution in a limit results in any of the following forms:

$$\frac{0}{0} \qquad \frac{\pm\infty}{\pm\infty} \qquad 0 \times \pm\infty \qquad \infty - \infty \qquad 1^\infty \qquad 0^0 \qquad \infty^0$$

we say the limit is in an **indeterminate form**.

Theorem (L'Hopital's Rule): Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval that contains a (except possibly at a) and. Suppose that $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is in the indeterminate form $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$., then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

provided that the limit on the right exists or is infinite.

Note: L'Hopital's rule still applies if a is replaced with infinity or negative infinity. It also applies to right and left hand limits.

Proof: (*special case only*)

Example 1: Find each of the following limits.

a) $\lim_{x \rightarrow 1} \frac{\ln(x)}{x-1}$

b) $\lim_{x \rightarrow \infty} \frac{\ln(x)}{x-1}$

Example 2: Determine each of the following limits.

a) $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$

b) $\lim_{x \rightarrow \infty} \frac{\ln(x)}{2\sqrt{x}}$

c) $\lim_{x \rightarrow \infty} \frac{e^{3x}}{x^2 + 3x + 5}$

Example 3: Determine what is wrong with each of the following limit calculations.

a) $\lim_{x \rightarrow \pi} \frac{\cos(x)}{x - \pi} \stackrel{H}{=} \lim_{x \rightarrow \pi} \frac{-\sin(x)}{1} = -\sin(\pi) = 0$

b) $\lim_{x \rightarrow 0} \frac{3x^2}{x^2 + 2x} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{6x}{2x + 2} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{6}{2} = 3$

Example 4: Find $\lim_{x \rightarrow 0^+} x \ln(x)$.

Example 5: Find $\lim_{x \rightarrow -\infty} x e^x$.

Example 6: Find $\lim_{x \rightarrow 0^+} \left(\frac{1}{\sin(x)} - \frac{1}{x} \right)$.

Dealing with Indeterminate Powers:

Thus far, the only indeterminate forms we have not addressed are the indeterminate powers. For this suppose that $\lim_{x \rightarrow a} f(x)^{g(x)}$ is of an indeterminate form (where $f(x)^{g(x)}$ is defined near a).

Example 7: Find $\lim_{x \rightarrow 0^+} x^x$.

Example 8: Find $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$.